NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

PRACTICE QUESTIONS

A cloud service provider schedules computational tasks on a single server, each taking 1 hour, with a profit (in credits) and deadline (in hours). The server can handle one task per hour, and tasks must be completed by their deadlines. Tasks: T1 (profit 200, deadline 3), T2 (150, 2), T3 (300, 4), T4 (180, 3), T5 (250, 5), T6 (220, 3). The challenge lies in tight deadlines (multiple tasks due at hour 3) and maximizing profit under contention.	A video streaming platform compresses subtitle data with 6 characters: 'A' (freq 20), 'B' (15), 'C' (30), 'D' (10), 'E' (25), 'F' (18). The challenge is generating efficient codes for real-time transmission with minimal bandwidth.
A data warehouse merges 6 sorted tables: sizes 15MB, 25MB, 10MB, 30MB, 20MB, 12MB. The challenge is minimizing computational cost with uneven sizes.	An investor allocates $2000 across 6 stocks, each with expected return (value) and cost (weight): Stock1 (400, 300), Stock2 (300, 200), Stock3 (500, 400), Stock4 (600, 500), Stock5 (350, 250), Stock6 (450, 350). The challenge is maximizing return with limited capital.
A conference center schedules events, each taking 1 day, with a booking fee (in dollars) and a deadline (days from now). With 7 events and tight deadlines, the challenge is to maximize revenue: Event1 (fee 500, deadline 4), Event2 (600, 3), Event3 (450, 2), Event4 (700, 5), Event5 (400, 2), Event6 (550, 4), Event7 (300, 1). Overlapping deadlines increase contention.	A cybersecurity system compresses log entries with 7 error codes: 'E1' (50), 'E2' (30), 'E3' (20), 'E4' (45), 'E5' (25), 'E6' (35), 'E7' (40). The challenge is minimizing storage for high-volume logs.
A backup system merges 7 sorted archive files: sizes 8GB, 12GB, 5GB, 20GB, 15GB, 10GB, 6GB. The	An air cargo plane with 1000kg capacity loads 5 goods: Good1 (value $800, 200kg), Good2 ($600, 150kg), Good3 ($1000,

challenge is handling large file sizes efficiently.	300kg), Good4 ($700, 250kg), Good5 ($900, 350kg). The challenge is maximizing value with strict weight limits.
A factory schedules 6 production jobs on a single machine, each taking 1 hour, with profit margins (in thousands) and deadlines (hours from now). Jobs: J1 (100, 2), J2 (150, 3), J3 (80, 1), J4 (120, 4), J5 (90, 2), J6 (200, 5). The challenge is optimizing under tight early deadlines and high-profit late jobs.	A music streaming service compresses MIDI note symbols: 'C' (40), 'D' (25), 'E' (30), 'F' (15), 'G' (20), 'A' (35). The challenge is reducing latency for real-time playback. Output the codes, compression ratio for 1000 notes (vs. 3-bit fixed encoding),
A relief agency allocates 500 units of capacity to 6 supply types: Food (200 relief points, 100 units), Water (150, 80), Medicine (300, 120), Shelter (250, 150), Clothing (180, 90), Blankets (220, 110). The challenge is maximizing relief under tight constraints	A bioinformatics lab merges 5 sorted DNA sequence files: sizes 100KB, 200KB, 150KB, 80KB, 120KB. The challenge is minimizing computational cost for large datasets.
An IoT network schedules 8 real-time data processing tasks, each taking 1 minute, with rewards (in priority points) and deadlines (minutes from now). Tasks: T1 (50, 2), T2 (70, 3), T3 (60, 1), T4 (90, 4), T5 (80, 2), T6 (100, 5), T7 (65, 3), T8 (55, 2). The challenge is handling a large number of tasks with overlapping deadlines.	A cloud storage system compresses metadata tags for files: 'Tag1' (60), 'Tag2' (40), 'Tag3' (20), 'Tag4' (50), 'Tag5' (30), 'Tag6' (25), 'Tag7' (45), 'Tag8' (35). The challenge is handling a large tag set for efficient storage. Output the codes, savings for 5000 tags (vs. 8-bit fixed), and time complexity.
A backup system merges 7 sorted archive files: sizes 8GB, 12GB, 5GB, 20GB, 15GB, 10GB, 6GB. The challenge is handling large file sizes efficiently.	A video conferencing system allocates 2000 Mbps bandwidth to 7 streams: Stream1 (600 priority, 400 Mbps), Stream2 (500, 300), Stream3 (800, 500), Stream4 (450, 250), Stream5 (700, 450), Stream6 (550, 350), Stream7 (400, 200). The challenge is maximizing quality under bandwidth limits.

Q. A CSP provider schedules computational tasks on a single server, each taking 1 hr, with a profit (in credits) & deadline (in hrs). The server can handle one task per hr & tasks must be completed by their deadlines.

Tasks: T1 (profit 200, deadline 3), T2 (150, 2), T3 (300, 4), T4 (180, 3), T5 (250, 5), T6 (220, 3).

The challenge lies in tight deadlines (multiple tasks due at hr 3) & maximizing profit under contention.

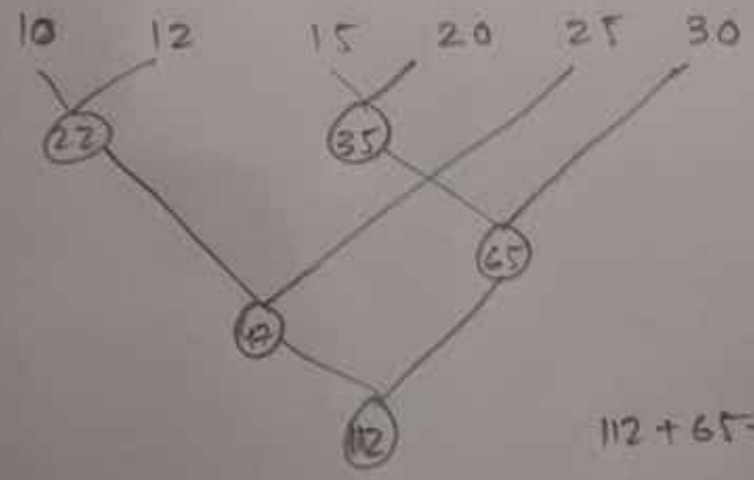
Tasks	T3	T5	T6	T1	T4	T2
Profit	300	250	220	200	180	150
Deadline	4	5	3	3	3	2

0 — 1 — 2 — 3 — 4 — 5

T4 T1 T6 T3 T5

Profit = 300 + 250 + 220 + 200 + 180 = $\boxed{1150}$.

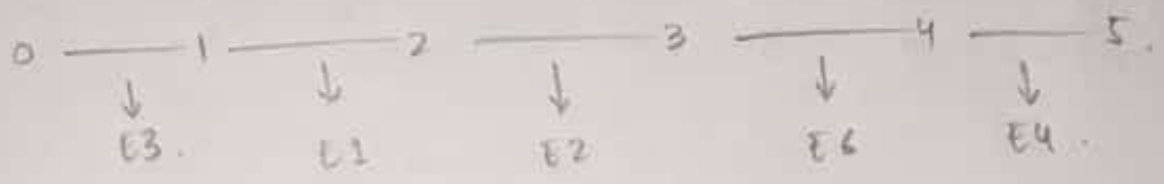
Q. A data warehouse merges 6 sorted tables: sizes 15MB, 25MB, 10MB, 30MB, 20MB, 12MB. The challenge is minimizing computational cost with uneven sizes.



112 + 65 + 47 + 35 + 22 = $\boxed{281}$.

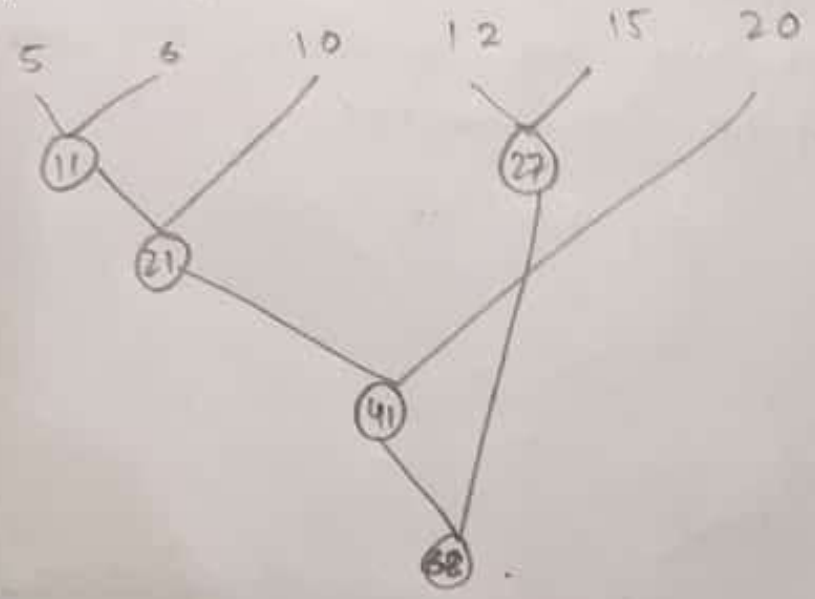
Q. A conference center schedules events, each taking 1 day, with a booking fee (in dollars) & a deadline (days from now). With 7 events & tight deadlines, the challenge is to maximize revenue: Event 1 (fee 500, deadline 4), Event 2 (600, 3), Event 3 (450, 2), Event 4 (700, 5), Event 5 (400, 2), Event 6 (550, 4), Event 7 (300, 1). Overlapping deadlines increase contention.

Events	E4	E2	E6	E1	E3	E5	E7
Fee	700	600	550	500	450	400	300
Deadlines	5	3	4	4	2	2	1



Profit = 450 + 500 + 600 + 550 + 700 = $\boxed{2800}$

Q. A backup system merges seven sorted archive files: sizes 8GB, 12GB, 5GB, 20GB, 15GB, 10GB, 6GB. The challenge is handling large file sizes efficiently.



68 + 41 + 27 + 21 + 11 = $\boxed{168}$

Q. A factory schedules 6 jobs on a single machine, each taking 1 hr, with profit margins (in thousands) & deadlines (hrs from now).

Jobs: J1 (100, 2), J2 (150, 3), J3 (80, 1), J4 (120, 4), J5 (90, 2), J6 (200, 5).

The challenge is optimizing under tight deadlines (early) & high-profit late jobs.

Jobs	J6	J2	J4	J1	J5	J3
Profit	200	150	120	100	90	80
Deadline	5	3	4	2	2	1

0 —— 1 —— 2 —— 3 —— 4 —— 5

↓ J5 ↓ J1 ↓ J2 ↓ J4 ↓ J6

Profit = 200 + 150 + 120 + 100 + 90 = $\boxed{660}$

Q. An IOT network schedules 8 real-time data processing tasks, each taking 1 minute, with rewards (in priority points) and deadlines (minutes from now). Tasks: T1 (50, 2), T2 (70, 3), T3 (60, 1), T4 (90, 4), T5 (80, 2), T6 (100, 5), T7 (65, 3), T8 (55, 2). The challenge is handling a large number of tasks with overlapping deadlines.

Tasks	T6	T4	T5	T2	T7	T3	T8	T1
Rewards	100	90	80	70	65	60	55	50
Deadlines	5	4	2	3	3	1	2	2

0 —— 1 —— 2 —— 3 —— 4 —— 5

↓ T7 ↓ T5 ↓ T2 ↓ T4 ↓ T6

Profit = 100 + 90 + 80 + 70 + 65 = $\boxed{400}$

Q. A relief agency allocates 500 units of capacity to 6 supply types: Food (200 relief points, 100 units), Water (150, 80), Medicine (300, 120), Shelter (250, 150), Clothing (180, 90), Blankets (220, 110). The challenge is maximizing relief under tight constraints.

Type	F	W	M	S	C	B
RP	200	150	300	250	180	220
Units	100	80	120	150	90	110
RP/Units	2	1.875	2.5	1.66	2	2

Type	M	F	C	B	W	S
RP	300	200	180	220	150	250
Units	200	100	90	110	80	150
RP/Units	2.5	2	2	2	1.875	1.66

Units = ((1×200) + (1×100) + (1×90) + (1×110) + (0×80) + (0×150)) = $\boxed{500 \text{ units}}$

RP = ((1×300) + (1×200) + (1×180) + (1×220)) = $\boxed{900}$

B	110 − 110 = 0
C	200 − 90 = 110
F	300 − 100 = 200
M	500 − 200 = 300

PRACTICE QUESTIONS.

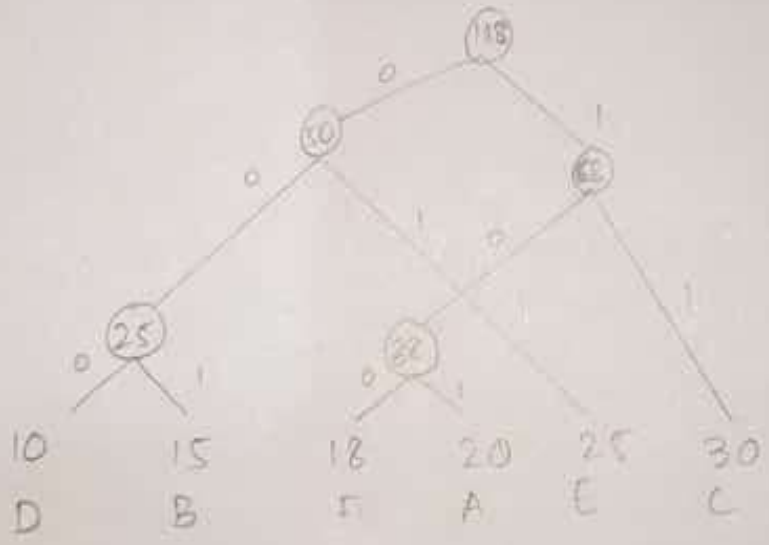
Q. A video streaming platform compresses subtitle data with 6 characterstics: 'A' (freq 20), 'B' (15), 'C' (30), 'D' (10), 'E' (25), 'F' (18). The challenge is generating efficient codes for real time transmission with minimal bandwith.

Char	A	B	C	D	E	F
Freq	20	15	30	10	25	18

$$\begin{array}{r} 118 \\ \times\ 8 \\ \hline 944 \end{array}$$ → Original size.

Sorting these in ascending order:

Char	D	B	F	A	E	C
Freq	10	15	18	20	25	30



A	101	3 × 20	60
B	001	3×15	45
C	11	2×30	60
D	000	3 × 10	30
E	01	2×25	50
F	100	3 × 18	54
			299

→ Compressed size

Q. An investor allocates \$2000 across 6 stocks, each with expected return (value) & cost (weight): Stock 1 (400, 300), Stock 2 (300, 200), Stock 3 (500, 400), Stock 4 (600, 500), Stock 5 (350, 250), Stock 6 (450, 350). The challenge is maximizing return with limited capital.

Stocks	St 1	St 2	St 3	St 4	St 5	St 6
Value	400	300	500	600	350	450
Weight	300	200	400	500	250	350
V/W	1.33	1.5	1.25	1.2	1.4	1.28

After sorting in descending order w.r.t V/W ratio

Stocks	St 2	St 5	St 1	St 6	St 3	St 4
Value	300	350	400	450	500	600
Weight	200	250	300	350	400	500
V/W	1.5	1.4	1.33	1.28	1.25	1.2

$$\text{Cost (Weight)} = ((1\times200)+(1\times250)+(1\times300)+(1\times350)+(1\times400)+(1\times500)) = 2000$$

$$\text{Return (Value)} = ((1\times300)+(1\times350)+(1\times400)+(1\times450)+(1\times500)+(1\times600)) = 2600$$

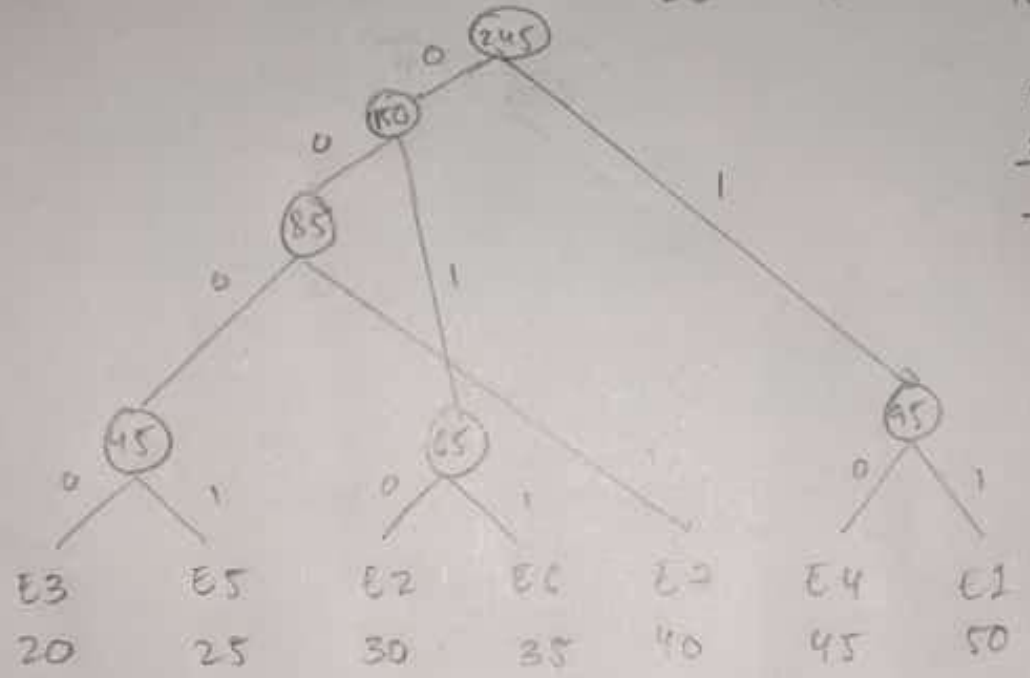
St 4	600 − 500 = 0
St 3	900 − 400 = 500
St 6	1250 − 350 = 900
St 1	1550 − 300 = 1250
St 5	1800 − 250 = 1550
St 2	2000 − 200 = 1800

$\boxed{\text{Cost (Weight)} = 2000}$

$\boxed{\text{Return (Value)} = 2600}$

Q. A cybersecurity system compresses log entries with 7 error codes : 'E1' (50), 'E2' (30), 'E3' (20), 'E4' (45), 'E5' (25), 'E6' (35), 'E7' (40). The challenge is minimizing storage for high-volume logs.

Codes	E1	E2	E3	E4	E5	E6	E7
freq.	50	30	20	45	25	35	40



245
× 8
1960 → Original size

E1	11	2 × 50	100
E2	010	3 × 30	90
E3	0000	4 × 20	80
E4	10	2 × 45	90
E5	0001	4 × 25	100
E6	011	3 × 35	105
E7	001	3 × 40	120
			$\boxed{625}$ → Compressed size.

Q. An air cargo plane with 1000kg capacity loads 5 goods: ~~Good1~~ Good 1 (value \$800, 200kg), Good 2 (\$600, 150kg), Good 3 (\$1000, 300kg), Good 4 (\$700, 250kg), Good 5 (\$900, 350kg). The challenge is maximizing value with strict weight limits.

Goods	G1	G2	G3	G4	G5
Value	800	600	1000	700	900
Weight	200	150	300	250	350
V/W	4	4	3.33	3	2.57

$$\text{Weight} = \left((1\times200) + (1\times150) + (1\times300) + (1\times250) + \left(\frac{100}{350}\times350\right)\right) = 1000$$

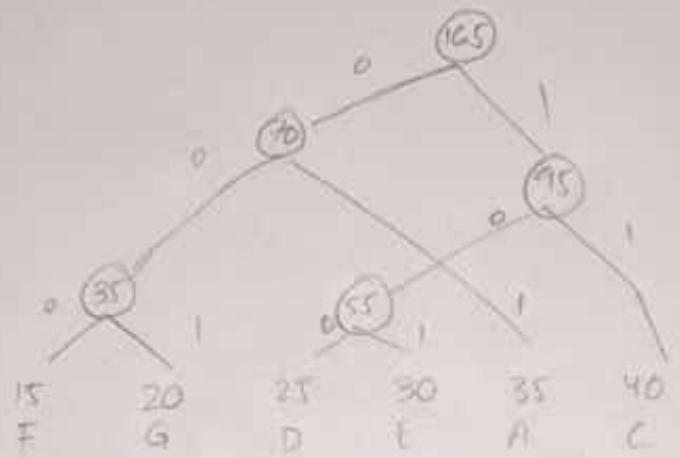
$$\text{Value} = \left((1\times800) + (1\times600) + (1\times1000) + (1\times700) + \left(\frac{100}{350}\times900\right)\right) = 3357.14$$

G5	$\frac{100}{350} = 0$
G4	$350 - 250 = 100$
G3	$650 - 300 = 350$
G2	$800 - 150 = 650$
G1	$1000 - 200 = 800$

Q. A music streaming service compresses MIDI note symbols 'C' (40), 'D' (25), 'E' (30), 'F' (15), 'G' (20), 'A' (35). The challenge is reducing latency for real time playback.

Output the codes, compression ratio for 1000 notes (vs 3-bit fixed encoding).

Char	C	D	E	F	G	A	
Freq	40	25	30	15	20	35	→ 165.



		freq × code length	
C	11	2 × 40	80
D	100	3 × 25	75
E	101	3 × 30	90
F	000	3 × 15	45
G	001	3 × 20	60
A	01	2 × 35	70
			420

• For avg bits per symbol :-

$$\text{Avg bits} = \frac{\text{freq of symbol} \times \text{length of symbol code}}{\text{total symbols}}$$

$$\text{Avg bits} = \frac{420}{165} = 2.545 \text{ bits/symbol.}$$

Compressed size = 1000 × 2.545 = $\boxed{2545 \text{ bits}}$

Original size = 1000 × 3 = 3000 (given)

$$\text{Compression Ratio} = \frac{3000}{2545} = \boxed{1.178}$$

$$\text{Compression ratio} = \frac{\text{Original size}}{\text{Compressed size}}$$

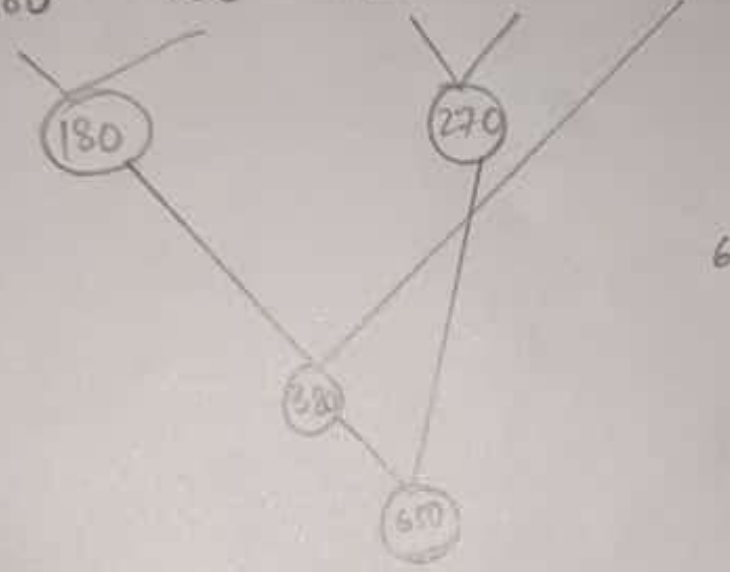
• For compressed size for 1000 bits :-

Compressed size = 1000 × (Avg bits per symbol).

P. A bioinformatics lab merges 5 sorted DNA sequence files: sizes 100KB, 200KB, 150KB, 80KB, 120KB. The challenge is minimizing computational cost for large datasets.

File	A	B	C	D	E
Size	100	200	150	80	120

File	D	A	E	C	B
Size	80	100	120	150	200



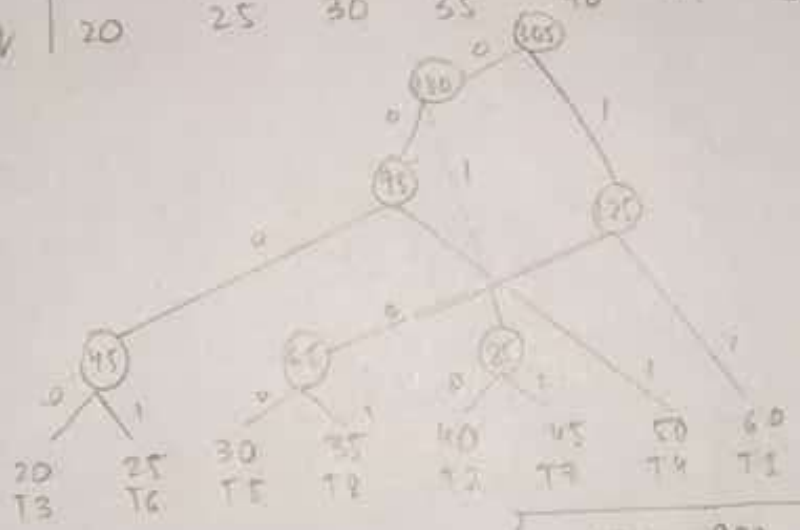
$650 + 380 + 270 + 180 = \boxed{1480}$

Q. A cloud storage system compresses metadata tags for files: 'Tag1' (60), 'Tag2' (40), 'Tag3' (20), 'Tag4' (50), 'Tag5' (30), 'Tag6' (25), 'Tag7' (45), 'Tag8' (35). The challenge is handling a large tag set for efficient storage.
Output the codes, savings for 5000 tags (vs, 8-bits fixed) & time complexity.

Tags	T1	T2	T3	T4	T5	T6	T7	T8
Freq	60	40	20	50	30	25	45	35

↳ 305

Tags	T3	T6	T5	T8	T2	T7	T4	T1
Freq	20	25	30	35	40	45	50	60



T1	11	2 × 60	120
T2	010	3 × 40	120
T3	0000	4 × 20	80
T4	001	3 × 50	150
T5	100	3 × 30	90
T6	0001	4 × 25	100
T7	011	3 × 45	135
T8	111	3 × 35	105
			900

Avg bits $= \frac{900}{305} = 2.950$

Compression size $= 5000 \times 2.950 = 14750$ bits

Original size $= 5000 \times 8 = 40000$ bits (given)

Compression ratio $= \frac{14750}{40000} = 0.368$

Savings $= 40000 - 14750 = \boxed{25250 \text{ bits}}$

Time complexity $= O(n \log n)$ → Min Heap

Q. A video conferencing system allocates 2000 Mbps bandwith to 7 streams Stream 1 (600 priority, 400 Mbps), Stream 2 (500, 300), Stream 3 (800, 500), Stream 4 (450, 250), Stream 5 (700, 400), Stream 6 (550, 350), Stream 7 (400, 200). The challenge is maximizing quality under bandwith limits.

Streams	St 1	St 2	St 3	St 4	St 5	St 6	St 7
Priority	600	500	800	400	700	550	400
Mbps	400	300	500	250	400	350	200.
P/M	1.5	1.66	1.6	1.6	1.75	1.57	2.

Streams	St 7	St 5	St 2	St 3	St 4	St 6	St 1
Priority	400	700	500	800	400	550	600
Mbps	200	400	300	500	250	350	400
P/M	2	1.75	1.66	1.6	1.6	1.57	1.5.

St 6	350 − 350 = 0.
St 4	600 − 250 = 350
St 3	1100 − 500 = 600
St 2	1400 − 300 = 1100
St 5	1800 − 400 = 1400
St 7	2000 − 200 = 1800

Mbps = ((1×200) + (1×400) + (1×300) + (1×500) + (1×250) + (1×350) + (0×400)) = $\boxed{2000 \text{ Mbps}}$.

Priority = ((1×400) + (1×700) + (1×500) + (1×800) + (1×400) + (1×550) + (0×600)) = $\boxed{3350}$.